Personal Meeting

October 26, 2026

1 Thresholding Derivation

1.1 Clarifications

1. λ **is real because minimization compares values.** Minimizing requires ordering objective values, and $\mathbb{C}$ has no ordering, so the objective $\frac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x})$ must be real; with $R(\mathbf{x}) \geq 0$ real, λ must be real. This is unrelated to the phase assumption $R(x) = r(|x|)$: λ would still be real for a phase-dependent penalty.

2. **The phase of $\hat{x}$ equals that of z because of a lower bound.** For fixed $|x|$ the penalty is fixed and $|x-z|^2 \geq (|x|-|z|)^2$; the bound is attained iff x has the phase of z, so the minimizer takes that phase. The polar form, the rectangular form and the triangle inequality all give this same bound and equality condition (Section 1.2, Phase).

1.2 Derivation Summary

Problem formulation.

$$\hat{\mathbf{x}} = \underset{\mathbf{x}\in\mathbb{C}^N}{\arg\min}\ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x})$$

- $\mathbf{z} \in \mathbb{C}^N$: observed complex image.
- $\mathbf{x} \in \mathbb{C}^N$: estimate.
- $R(\mathbf{x})$: sparsity-promoting penalty, $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$ or $\|\mathbf{x}\|_{1/2}^{1/2}$.
- $\lambda > 0$: weight between the data term and the penalty; real, since the objective must be real-valued to be minimized.

Separability. If $R(\mathbf{x}) = \sum_i R_i(x_i)$,

$$\begin{aligned}\min_{\mathbf{x}\in\mathbb{C}^N}\ \tfrac{1}{2}\|\mathbf{x}-\mathbf{z}\|_2^2 + \lambda R(\mathbf{x}) &= \min_{\mathbf{x}\in\mathbb{C}^N}\ \sum_{i=1}^{N}\Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big] \\ &= \sum_{i=1}^{N}\ \min_{x_i\in\mathbb{C}}\ \Big[\tfrac{1}{2}|x_i - z_i|^2 + \lambda R_i(x_i)\Big],\end{aligned}$$

so each $\hat{x}_i$ is found independently. Below, the index i is dropped ($x = x_i$, $z = z_i$, $R = R_i$):

$$\hat{x} = \underset{x\in\mathbb{C}}{\arg\min}\ \tfrac{1}{2}|x-z|^2 + \lambda R(x).$$

Phase. If $R(x) = r(|x|)$ (true for $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_0$, $\|\mathbf{x}\|_{1/2}^{1/2}$, defined through $|x_i|$), then for fixed $|x|$ the penalty is constant and $|x-z|^2 \geq (|x|-|z|)^2$, with equality iff x has the phase of z:

- Polar, $x = \rho e^{j\theta}$, $z = me^{j\varphi}$: $|x-z|^2 = \rho^2 + m^2 - 2\rho m\cos(\theta-\varphi) \geq (\rho-m)^2$, "=" iff $\theta = \varphi$.
- Rectangular, $x = a + jb$, $z = c + jd$: $|x-z|^2 = |x|^2 + |z|^2 - 2(ac+bd) \geq (|x|-|z|)^2$, since $ac + bd \leq |x||z|$ (Cauchy–Schwarz), "=" iff $(a,b) = \kappa(c,d)$, $\kappa \geq 0$.
- Triangle inequality: $|x-z| \geq \big||x|-|z|\big|$, "=" iff $x = \kappa z$, $\kappa \geq 0$.

Hence, with $\rho = |x|$ and $m = |z|$,

$$g(\rho) = \tfrac{1}{2}\big(|x|-|z|\big)^2 + \lambda r(|x|) = \tfrac{1}{2}(\rho-m)^2 + \lambda r(\rho), \qquad \hat{\rho} = \underset{\rho\geq 0}{\arg\min}\ g(\rho), \qquad \hat{x} = \operatorname{sgn}(z)\,\hat{\rho}.$$

ℓ_1: soft thresholding. $r(\rho) = \rho$, $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\rho$, $g'(\rho) = \rho - m + \lambda$:

- $m > \lambda$: $g'(\rho) = 0 \Rightarrow \hat{\rho} = m - \lambda$.
- $m \leq \lambda$: $g'(\rho) > 0$ for $\rho > 0 \Rightarrow \hat{\rho} = 0$.

$$\mathcal{S}_\lambda(z) = \operatorname{sgn}(z)\, \max(|z| - \lambda,\, 0).$$

ℓ_0: hard thresholding. $r(\rho) = \mathbf{1}[\rho > 0]$, $g(\rho) = \frac{1}{2}(\rho - m)^2 + \lambda\, \mathbf{1}[\rho > 0]$:

- $\rho = 0$: $g(0) = \frac{1}{2}m^2$.
- $\rho > 0$: best $\rho = m$, $g(m) = \lambda$.
- $\hat{\rho} = m$ iff $\lambda < \frac{1}{2}m^2$, i.e. $m > \sqrt{2\lambda}$ ($\hat{\rho} = 0$ at equality).

$$\mathcal{H}_{\sqrt{2\lambda}}(z) = z \cdot \mathbf{1}\big[|z| > \sqrt{2\lambda}\big].$$